Is mass a contained rhythm, and time the same rhythm read? Eleven registered translations confronted with MICROSCOPE, lunar laser ranging, interferometry and the Margolus–Levitin bound

Ugo Lanciano
Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635

Version 2.64.1, 2026-10-25. Archived on Zenodo, https://doi.org/10.5281/zenodo.23643718. Licence CC BY 4.0. Developed with the help of AI assistants, as declared in the record.

Abstract

Physics already ties mass to time: every mass carries a rhythm, the Compton frequency mc^2/h, and all bodies fall in the same way within a few parts in 10^{15}. This deposit tests one thesis of the framework: the same counting is time when it is read, and mass when it is contained directly in the matrix of the point of observation . Read: it becomes a row of distinct events, like the ticks of a clock. Contained, in the reading chosen by the AI for the calculations: it stays inside the thing as a rhythm that produces no distinct events, like a top spinning in a closed box; in this reading mass is made of time, time held inside the thing. In the parametrisations compared, the data exclude that the count of the pieces weighs and are compatible with the value , the total energy including binding. It is the twin of deposit 2.62 on observed objects (DOI 10.5281/zenodo.23609967 , corrected in version 2.62.1): in the counting, fractions give objects and distances, multiples give time and mass.

Ugo Lanciano, October 2026. Licence CC BY 4.0 (texts); code MIT. The same text in Italian is in LEGGIMI_IT.md.

Version 2.64.1 (25 October 2026): corrections and two new sections. An independent check requested by the author found places where version 2.64 presented as a result about the thesis what depended on translation choices made by the AI and not declared. This version corrects them and adds section 9, on time with a direction, and section 10, on mass in the matrix of whoever observes; the programs, the calculations and the numbers of 2.64 do not change.

1. The author speaks of a multiple «contained directly in the matrix of the point of observation». Version 2.64 had read it as a rhythm closed inside the object, and all eleven translations are properties of the body: this is a choice of the AI. The results hold for these readings; the translation using the matrix of whoever observes is item M-i of the new section 10, and it coincides with standard physics.
2. The distinction between count and value was a reading by the AI of a premise of the author: in the parametrisations compared the data exclude the count and are compatible with the value, but the premise does not choose between the two readings.
3. Item M-h (masses growing with the age of the cosmos) is the AI's, and so is the start of the count at the beginning of the cosmos.
4. Reading the iron nucleus with the author's sentence on the average GCD is a choice of the AI; the condition on the average GCD is not tested.
5. That possible copies of the Compton rhythm (M-g) would remain an internal record of the particle was a hypothesis of the AI, not tested; the separation from the observed objects holds in the static model of deposit 2.62.
6. The number of copies emitted by a source is a datum distinct from the mass and is not derived from it: it depends on the conditions of the source, among them the illumination and the temperature.
7. New section 9, «Physical time and time with a direction»: read as the increase of the relations in the observer's record, the direction of time is a quantity different from the direction of heat, in agreement at some moments and opposite at others; no new prediction.
8. New section 10, «Mass in the matrix of whoever observes»: the missing translation (item M-i). Read as the part of the observer's copies that stays the same at every tick, mass coincides with standard physics and in no case examined gives a different number.

Details: 04_controllo_262_264/PIANO_CORREZIONI_v1.md (in Italian, as registered); the work on time is in the folder 03_direzione_tempo, the work on item M-i in the folder 06_massa_matrice_osservatore.

This work and deposit 2.62 are the two halves of a single step of the author's framework, the one farthest from the usual way of thinking about physics: in the counting, fractions give the observed objects and their distances (deposit on

objects, 2.62), and multiples give time and mass (this deposit). Each can be read on its own.

In brief

This work puts a precise idea to the test: **mass and time are not two separate things, but the same rhythm in two different conditions.**

Physics already knows two fundamental facts about mass. First: every mass carries an extremely fast internal rhythm, the Compton frequency mc^2/h, which for an electron is about $1.24 \cdot 10^{20}$ cycles per second. Second: in a gravitational field all bodies fall in the same way, whatever their composition, with a precision verified down to a few parts in 10^{15}.

The author's thesis joins these facts: **the same counting is time when it is read, and mass when it is contained directly in the matrix of the point of observation.** In this work «contained» was read as a rhythm closed inside the object: this is a choice of the AI, not the only possible one.

- **Read (time):** the rhythm comes out and becomes a row of distinct events, one after another, like the ticks of a clock marking the passing of time.
- **Contained (mass), in the AI's reading:** the same rhythm stays closed inside the object without producing events visible from outside, like a top spinning inside a closed box without leaving marks. When the rhythm stays closed in this way, it shows itself as mass.

In this reading, **mass is made of time: it is time held inside the thing.**

In the parametrisations compared, the data exclude that what produces weight is the **count** of the single pieces that make up the body, and they are compatible with the overall **value**, that is, the total energy enclosed, including the binding energy that holds the pieces together.

To test the thesis we translated it in eleven different ways, each with a mathematical definition, and registered all the translations before doing the calculations, together with a reference (standard physics) and a threshold for neutrinos. Then we compared them with measurements. The main results are four.

- **In the parametrisations compared, the value weighs, not the count.** If the count of the pieces were what weighs, titanium and platinum would fall with a difference of 7.5 parts in 10 000; MICROSCOPE sees no difference beyond a few parts in 10^{15}.
- **A mass at rest produces no distinct events.** Its state changes only by a phase. The same energy, if released, can produce at most two distinct states per Compton cycle (Margolus and Levitin 1998).
- **Whether the Compton rhythm leaves readable copies remains open.** The data at hand are not enough to decide.
- **Today no translation gives a testable prediction that differs from standard physics.** The translations that hold coincide with $E = mc^2$ and with relativity; the only one that departs from them (the checkerboard of deposit 2.40) gives an effect of 10^{-42}, beyond the reach of any planned experiment. All eleven translations read «contained» as a property of the body. The reading with the matrix of whoever observes is item M-i (§10, new in this version): mass as the part of the observer's copies that stays the same at every tick. It too coincides with standard physics.

Physical time and time with a direction (§9, new in this version). The author's framework ties the direction of time to the increase in the number of relations, linked in a network, inside the matrix of whoever observes. Read in this way, the direction of time and the direction of heat are two distinct quantities: in standard quantum models, reconstructed from published experiments, the relations in the observer's record are at some moments above their initial value while heat flows from cold to hot, and at others below it while heat flows in the normal direction. The relations between the observed parts, instead, follow the heat through a known identity. No new prediction.

How to read this text. Section 2 explains the thesis without formulas and should be read first; §2.3 shows it with three everyday examples. Section 1 summarises what physics already knows: on a first reading it can be skimmed. Section 3 says how the thesis was tested; §4 gives the results, starting with three calculations that can be done by hand. Sections 9 and 10, new in this version, deal with time with a direction and with mass in the matrix of whoever observes; they can be read on their own. Technical words are explained in brackets the first time they appear, and again in the glossary at the end. Numbers such as 2.25, 2.36.2, 2.38, 2.40 and 2.62 refer to earlier works of the author deposited on Zenodo; the list with DOIs is at the end of the references.

1. What is already known

Mass and energy. The mass of a body is its total energy, measured where the body is at rest, divided by c^2 ($E = mc^2$). All energies count: that of the pieces, that with which they move inside the body, and the negative one that binds them. This is why a nucleus weighs less than its separate pieces: the 26 protons (with their electrons) and the 30 neutrons of iron-56, taken one by one, weigh 0.945 % more than the iron atom (AME2020 tables). And for the same reason a box of mirrors containing one joule of light weighs $1.1 \cdot 10^{-17}$ kg more than the empty box.

Mass that comes from energy. More than 99 % of the mass of ordinary matter is in protons and neutrons, and about 95 % of their mass comes from the energy of the quarks and gluons enclosed in them, not from the quark masses. The lattice calculation of Dürr and others (2008) recovers the nucleon mass: 0.936 GeV, with an uncertainty of about 0.03 GeV, against 0.939 measured, starting from only three measured masses: those of the pion and the kaon, which fix the masses of the light and strange quarks, and that of the Ξ particle, which fixes the scale (fixing the scale with the Ω particle gives 0.953 GeV).

The Compton rhythm. To every mass corresponds a frequency, $\nu = mc^2/h$: $1.24 \cdot 10^{20}$ cycles per second for the electron, $2.27 \cdot 10^{23}$ for the neutron. Lan and others (2013) used it to build a clock tied to the mass of an atom. Whether this rhythm can be «seen» on its own is debated: Wolf and others (2011) show that in atom interferometers the difference between the «Compton phases» of the two paths is zero in general relativity and in a wide class of theories, and that those experiments measure the fall, not the rhythm.

Universal free fall. All bodies fall in the same way: this is the equivalence principle. The most precise measurement is that of MICROSCOPE (Touboul and others 2022): between an alloy of titanium and one of platinum the relative difference of the accelerations is $\eta = (-1.5 \pm 2.3 \pm 1.5) \cdot 10^{-15}$. Lunar laser ranging, which measures the distance of the Moon to the millimetre, says that the Earth and the Moon fall towards the Sun in the same way within $(-2.1 \pm 2.4) \cdot 10^{-14}$, and that the mass that attracts and the mass that is attracted coincide for aluminium and iron within $3.9 \cdot 10^{-14}$ (Singh and others 2023). Atoms placed in a superposition of two internal states fall like the others within a few parts in 10^9 (Rosi and others 2017). Projects under study aim at 10^{-17}.

Gravity and clocks. Near a mass clocks run slower, all in the same proportion: this was seen with a hydrogen maser carried to an altitude of 10 000 km (within one part in 10^4) and with two optical clocks 33 cm apart in height.

Mass, time and distinct states. Margolus and Levitin (1998) proved that a system with mean energy E, counted from its ground state (the state of lowest energy), cannot pass through distinct states (technically, orthogonal ones: perfectly distinguishable) faster than a certain rate: in a long sequence, at most 2E/h per second. Lloyd (2000) turned this into a limit on computation: one kilogram of matter, if all its energy mc^2 were available, could perform at most $5.43 \cdot 10^{50}$ logical operations per second. The decisive point is the zero of energy: a system in a state of definite energy (an eigenstate) passes through no distinct state; its state changes only by a phase, like a hand that turns while nothing observable changes.

Clocks and interference. If a particle carries a true internal clock (for example two internal states that oscillate) and goes along two paths with different proper times, the clock records which path it took and the interference fringes lose contrast (Zych and others 2011). Margalit and others (2015) showed this with a clock made of two spin states, and recall that in ordinary interferometry, without an internal clock, «time cannot yield "which path" information». For a body made of many oscillating parts, the difference in proper time is enough to make interference disappear: for one gram at room temperature in two positions one micron apart, in about a thousandth of a second (Pikovski and others 2015).

Feynman's checkerboard. A particle can be described as a path on a checkerboard that zigzags along the directions of light rays; mass is how often the path turns. This construction, known as the Dirac quantum walk (Bialynicki-Birula 1994; Meyer 1996; Arrighi, Nesme and Forets 2014; D'Ariano and Perinotti 2014), was put to the test in deposit 2.40.

Neutrinos. JUNO (2026) measured the differences between the squares of the neutrino masses to one per cent and favours the normal ordering (the lightest is the first) at at least 2.1σ. DESI, together with the cosmic background and within the standard cosmological model, limits the sum of the masses to less than 0.064 eV; with a method that takes into account that masses cannot be negative, the limit drops to 0.053 eV, a value even below the minimum imposed by oscillations (0.0587 eV), as the DESI collaboration itself points out.

2. The thesis, motivated and described

2.1 The framework, in a few lines

In the author's framework the local world arises from a relation that is counted through a common unit, its greatest common divisor (GCD). An example with small numbers: to compare two quantities such as 3 and 5 one needs their largest common measure, which here is 1 (the greatest common divisor). To measure both, this unit must be repeated three times in the first and five times in the second. Counting thus produces **copies** of the unit. Whoever is inside the world, like us, arranges them in two ways:

- **in a row**, one after another: these are the **multiples**, and they appear as time;
- **side by side**, all together: these are the **fractions**, and they appear as space.

The set of copies is the **matrix**. The point of observation of whoever looks is a part of this matrix.

2.2 What the thesis says

In the author's words: «The same multiple, when it is contained directly in the matrix of the point of observation, turns into mass, that is, into concrete matter as we usually understand it.»

The author speaks of a multiple contained in the matrix of the point of observation, that is, in whoever observes. In this work we read it, by a choice of the AI, as a counting in a row that stays closed inside an object without showing itself outside: in this reading concrete matter is made of contained time. All the translations of §3 are properties of the body; the translation using the matrix of whoever observes, item M-i, is in §10.

Said in the words of physics, in the AI's reading:

- **the read multiple is time.** It is a row of distinct states, one after another, that an observer can count: the ticks of a clock;
- **the contained multiple is mass.** It is the same counting when it stays inside the thing: a rhythm that produces no distinct states, that changes the thing only by a phase. In physics this rhythm has a name, the Compton frequency;
- **what weighs is the value of the rhythm, not the count of the pieces.** The value is the total energy of the thing, binding included. The count of the pieces is the sum of the masses of the pieces taken one by one, without the binding.

Four words, defined at once:

- **multiple:** counting forward, one copy after another;
- **read:** becoming a sequence of distinct states, observable as time passing;
- **contained:** in the AI's reading, remaining an internal rhythm, a phase, without producing distinct states (for the author: contained directly in the matrix of the point of observation);
- **value:** the total energy contained, which relativity calls the rest mass multiplied by c^2.

2.3 Three examples

The closed top. A top spins inside a sealed box. From outside one cannot count how many turns it makes: it leaves no marks. Yet the box weighs more: for 100 joules of rotational energy, $1.1 \cdot 10^{-15}$ kg more (this is known physics, $E = mc^2$). If the top hits the wall, its turns become events: a noise, a little heat. Contained, the rhythm weighs; read, it becomes events in time.

Light in a box. A ray of free light has no mass: it has the frequency of its wave, but no rhythm at rest, that is, no Compton frequency; along its path proper time is zero, and it always travels at the speed of light. Closed in a box of mirrors, the same light gives the box a mass: one joule of light, $1.1 \cdot 10^{-17}$ kg. The same energy, contained, becomes mass.

The iron nucleus. The pieces of iron-56, separated, weigh 0.945 % more than the nucleus. When the pieces join, part of their value leaves as radiation: it becomes read, that is, events that flow in time, and so the mass that stays contained decreases. In the reading of the thesis: less rhythm closed inside, more events released outside. A sentence of the author goes in the same direction: if, coming closer, the average GCD decreases, the total energy decreases too (exact words in the background file). Reading the iron nucleus with this sentence is a choice of the AI: the author's condition on the average GCD was not tested. What weighs is what remains: the value, not the number of pieces.

2.4 Why think of it this way

1. **Physics already ties mass and time together.** A mass is a rhythm (Compton); near a mass all rhythms slow down together. The thesis does not add a new link: it says that mass and time are the same counting in two conditions.
2. **A single rule separates the two.** Read or contained. The rule introduces no new constant: units pass through h and c, as in known physics.
3. **The bridge between the two conditions is a known theorem.** The Margolus–Levitin bound says how many distinct states per second a given energy can produce at most. Read with the thesis: the contained multiple is a store of possible time, which becomes time when it is released. It is the «mass as frozen time» of deposit 2.25.
4. **Consistency with the starting framework.** In the framework the multiple is mass, and mass and time are two faces of the same counting. The author's words are in the background file, kept apart from this text.

2.5 How it reads some known facts

- **$E = mc^2$.** The value of the contained rhythm is the energy: it is the same equality, read as «what is contained weighs».
- **The mass defect.** Binding makes part of the value leave as radiation: what stays contained is worth less than the sum of the pieces (iron, §2.3).
- **Massless light and the box of light.** What has no rhythm at rest (no Compton frequency) has no mass; the same energy closed in a box does (§2.3). The author's book says it like this: what has no weight, like light, is «what does not beat».
- **The proton mass.** About 95 % is energy of the quarks and gluons enclosed inside: contained energy, hence mass.
- **Universal free fall.** If what weighs is the value, that is, all the contained energy, every body falls in the same way whatever its composition: this is the equivalence principle. The author's book says that near a mass the shift of rhythms «rescales together, in the same proportion, all the rhythms readable from that point».
- **Slower clocks near a mass.** Near a large mass, seen from far away, the time of clocks (the read rhythm) slows down, and the energy of a body (the contained rhythm) is worth less, exactly in the same proportion. On the spot, instead, the measured mass of that body does not change. In the reading of the thesis, gravity, mass and time move together because they are the same rhythm; in the AI's reading it is also the author's sentence of §2.3 (if, coming closer, the average GCD decreases, the energy decreases) read for gravity; the condition on the average GCD is not tested.

2.6 What it costs, said plainly

- **Mass is not a substance.** A mass at rest does nothing distinguishable: its state changes only by a phase. It looks like a paradox, but it is what quantum mechanics says of every state of definite energy.
- **The value of a single mass is a datum.** The thesis does not say how much the electron or the proton weighs: it takes it from measurements, as standard physics does. In the author's framework the value of each mass depends on how the matrix is built, and cannot be derived (background file).
- **Today it gives no new numbers.** The translations that hold coincide with standard physics (§4).
- **One question remains open:** whether the Compton rhythm leaves readable copies (§4.4).

2.7 What is new with respect to the closest studies

We did not carry out a systematic search of the literature; the closest works we know are these.

- **Margolus and Levitin (1998), Lloyd (2000)** tie energy (hence mass) to the maximum number of distinct states per second. Here that link becomes the bridge between the two conditions of counting, read and contained.
- **Zych and others (2011)** distinguish the phase accumulated by a particle from the internal clock that records the path, and propose to test the hypothesis that proper time is a quantum degree of freedom. Here the same distinction separates the translations: those in which the Compton rhythm remains a phase and the one in which it would leave copies.
- **Wolf and others (2011)** discuss whether the Compton rhythm is observed in interferometers; **Lan and others (2013)** make a clock of it; **Greenberger (2010)** discusses the conceptual problems that tie time and mass in quantum mechanics.
- **Pikovski and others (2015), Zych and Brukner (2018)** treat the mass of a composite body as its internal energy in relativity and quantum mechanics: the same thing that here is called the value.
- **Dirac quantum walks** give mass as a turning amplitude on a lattice (deposit 2.40 is a case of it).

The step that belongs to this work is another: to take the thesis «the contained multiple is mass», translate it in eleven ways (all as properties of the body), register them before the calculations and let the data choose. The data exclude that

the count of the pieces weighs and are compatible with the value; whether the contained rhythm remains only a phase or also leaves readable copies, the data at hand do not decide (§4.4).

---

3. How the thesis was tested

3.1 The thirteen items

In the test of 2.64 there are thirteen items (version 2.64.1 adds item M-i, in §10): eleven translations, that is, mathematical definitions of «contained multiple = mass», plus the reference M-0 (standard physics) and the threshold N for neutrinos. All were written and registered before the calculations, with the expected outcome written next to them.

item	what it says, in short
M-0	no consequence beyond standard physics (reference)
M-a	mass is the rhythm of the multiple in proper time (Compton), with the postulate of deposit 2.25 (phase of the count = action/$\hbar$)
M-b1	mass is the turning angle per step of the checkerboard (deposit 2.40); the walkers are nucleons and electrons; the step is the Planck time
M-b2	as M-b1, but a whole body is a single walker
M-c	mass is how much the body slows the rhythms in the space around it, summed over a surface enclosing it (Komar mass)
M-d1	mass is the number of copies of the source (premise P4 of deposit 2.36.2), and the copies are the pieces counted without binding
M-d2	as M-d1, but the copies carry the value: the total energy, binding included
M-e	mass and the slowing of time (a premise of the author, in the background file): seen from far away, the value of a mass near another drops in the same measure in which clocks slow down
M-f1	mass is the number of distinct states per second, counting the energy from the state at rest
M-f2	mass is the ceiling on the number of distinct states per second, counting the energy from the vacuum
M-g	the Compton rhythm leaves a readable copy at every cycle
M-h	(an item of the AI, with the start of the count at the beginning of the cosmos chosen by the AI) mass is the number of multiples accumulated since the beginning of the cosmos, and grows with its age
N	not a translation: the decision threshold for the neutrinos of deposit 2.38

3.2 The data

MICROSCOPE (titanium and platinum); lunar laser ranging (aluminium and iron; Earth and Moon; the constancy of G in time, $(-5.0 \pm 9.6)\cdot 10^{-15}$ per year); a neutron interferometer with arms at different heights (separation between the arms of about one centimetre and coherence time of about 10^{-4} s, as reported by Zych and others 2011); the AME2020 atomic masses; the Margolus–Levitin and Lloyd bounds; JUNO 2026 and DESI for neutrinos. The physical constants are those of the Particle Data Group (2025). The natural isotopic abundances, the radius and mass of the Earth and the Moon, and the age of the cosmos (13.8 billion years) are known values, taken without a downloaded source and declared: no outcome depends on them beyond a few per cent.

3.3 The criteria, fixed in advance

Seven criteria, the same for all items: $E = mc^2$ and the mass defect; universal free fall within MICROSCOPE; mass that comes from energy (the nucleon at 95 %); no new constant beyond h, c and G; no mass value derived by fitting; relativity (clocks in motion and in gravity); the limiting cases (no multiple, no mass; light without mass; a large mass fixed in the matrix).

The outcome rules, also fixed in advance: **excluded** if the prediction is more than three standard deviations from the datum (or exceeds the published limit); **compatible** if it is within two; **not testable today** if the predicted effect is at least ten times below today's sensitivity; **distinctive** if the effect exceeds the sensitivity of an experiment existing or planned by 2035 without being excluded by the data; **not conforming** when a criterion cannot be checked because the translation does not define something that is needed.

---

4. What comes out

4.1 Three calculations that can be done by hand

The count and the value. One takes the mass of each atom of the two MICROSCOPE alloys and compares it with the sum of the masses of its pieces taken one by one. For the titanium alloy the count of the pieces exceeds the value by 0.9342 %; for the platinum alloy by 0.8585 %. If the count of the pieces were what weighs, the two alloys would fall with a discrepancy equal to about the difference between these two numbers: 7.5 parts in 10 000. MICROSCOPE measures a discrepancy compatible with zero, with an error of 2.75 parts in 10^{15}.

The copies of the Compton rhythm. In a neutron interferometer the two arms are at different heights, and the higher one has a proper time that flows a little faster. With a height difference of one centimetre for 10^{-4} s, the difference in proper time is $1.09 \cdot 10^{-22}$ s, and the Compton period of the neutron is $4.40 \cdot 10^{-24}$ s: about 25 cycles of difference. If each cycle left a readable copy, the two arms would carry records differing by 25 copies. The calculation is simple: $\Delta N = m\, g\, \Delta h\, t / h \approx 24.8$.

One kilogram. The energy of one kilogram is $mc^2 \approx 9 \cdot 10^{16}$ J. Lloyd's bound gives at most $5.43 \cdot 10^{50}$ operations per second: four logical operations per Compton cycle. For a long sequence of distinct states, Margolus and Levitin give half of that: two states per cycle.

4.2 In the parametrisations compared: count excluded, value compatible (M-d1, M-d2)

The translation «the copies are the pieces counted without binding» (M-d1) predicts a difference in fall between titanium and platinum $\eta = 7.5 \cdot 10^{-4}$: it is $2.7 \cdot 10^{11}$ times the error of MICROSCOPE. At most one part in $2 \cdot 10^{11}$ of gravity ($\varepsilon \leq 5.3 \cdot 10^{-12}$) could act on the count of the pieces instead of on the value. Counting nucleons instead of the masses of the pieces gives the same order ($\eta = 8.0 \cdot 10^{-4}$). The same translation applied to the mass that attracts, aluminium against iron, exceeds the lunar-laser limit by $1.3 \cdot 10^{10}$ times. Applied to the gravitational binding of the Earth and the Moon (estimated with uniform spheres: $4.2 \cdot 10^{-10}$ of the Earth's energy and $1.9 \cdot 10^{-11}$ of the Moon's), it predicts that the Earth and the Moon fall towards the Sun with a difference of $4.0 \cdot 10^{-10}$, against a measurement of $(-2.1 \pm 2.4) \cdot 10^{-14}$: $1.7 \cdot 10^{4}$ standard deviations. These are exclusions of a parametrisation within the models with which those measurements are read, not separate measurements of each mass.

The translation «the copies carry the value» (M-d2) is compatible with everything: it is standard physics. In the parametrisations compared, therefore, the data exclude that the sum of the masses of the pieces weighs and are compatible with the value (the total energy, binding included). In September 2026 we had read a premise of the author (what passes from one register to another is only the partial premise of the new history, which is formed with the present register; exact words in the background file) as the distinction between the count, which is transmitted, and the value, which is formed in the register that receives. That reading is the AI's: the premise does not by itself choose between count and value, and the data do not test it.

4.3 The contained multiple produces no distinct states (M-f1, M-f2)

For a particle at rest in a state of definite energy, the overlap between the initial state and the state after any time is always 1 in modulus: the state changes only by a phase. The translation «mass is the number of distinct states per second» (M-f1) would therefore give zero mass to an electron at rest, and it falls by its own definition.

The reading as a ceiling (M-f2) holds instead: counting the energy from the vacuum, the same energy mc^2, if it becomes evolution, allows at most two distinct states per Compton cycle in a long sequence, four logical operations per cycle in Lloyd's count. Together, the two items give the AI's reading in precise form: **the contained multiple, read as a rhythm inside the body, is mass and produces no distinct states; when it is released it becomes time, with a maximum rate fixed by the mass itself.**

4.4 Does the Compton rhythm leave copies? (M-g)

If at every Compton cycle matter left a readable copy, in the neutron interferometer of §4.1 the two arms would carry records differing by about 25 copies: the records could be distinguished and the fringes would disappear. Fringes of neutron interferometers in gravity have been observed since 1975 (Colella, Overhauser and Werner). But this is not enough to close the question, for two reasons: the contrast measurement in the geometry used is not among the sources of this work, and the sources do not agree on how to read the observed gravitational phase (for Zych and others, in their scheme, it is the phase difference due to proper time; for Wolf and others, in atom interferometers the difference of the Compton phases between the paths is zero). Zych and others present the nearby hypothesis, that proper time is a quantum degree of freedom, as a possibility to be tested. **Outcome: not concluded.** It remains firm that in standard quantum mechanics a phase that changes in the same way across the whole state gives no information about the path (Margalit and others 2015). A neutron in its ground state has no internal degree of freedom that changes state at every Compton cycle: for a readable copy to remain, an extra degree of freedom would be needed, as in the hypothesis of a quantum proper time of Zych and others. M-g, therefore, does not follow from standard physics: it remains not concluded as a hypothesis that goes beyond it. To decide, the original 1975 source and an analysis of proper time in the two arms are needed.

4.5 The checkerboard of deposit 2.40 (M-b1, M-b2)

On the checkerboard mass is the turning angle per step, and deposit 2.40 has already shown that the fall depends slightly on mass: $a = -g\,\theta/\tan\theta$, that is, a correction of about $\theta^2/3$. For a body made of many walkers two rules are needed that deposit 2.40 does not give, and that here are chosen and declared: the walkers are nucleons and electrons, and the binding energy falls without correction, like a massless walker. With the step equal to the Planck time, titanium and platinum would fall with a difference of about $1.1\cdot10^{-42}$ (with a three-dimensional estimate, $4.5\cdot10^{-42}$): twenty-five orders of magnitude below the future projects at 10^{-17}. **Outcome of M-b1: not testable today**, and not conforming to one criterion (deposit 2.40 does not show that massless walkers bound together form a body with mass). If the step were not the Planck one but free, MICROSCOPE would require a step shorter than about 10^{-21} m (an energy of the step above $1.5\cdot10^5$ GeV; $3.1\cdot10^5$ GeV with the three-dimensional estimate), using the widest end of the two-standard-deviation interval.

A whole body as a single walker (M-b2) would have a turning angle of millions of radians. But on the checkerboard an angle that exceeds a full turn (360 degrees) starts again from the beginning: the checkerboard does not distinguish that body from a much lighter walker, and so does not assign it a unique mass. **Outcome of M-b2: not conforming.**

4.6 Masses that grow with the age of the cosmos (M-h)

Item M-h is the AI's, and starting the count at the beginning of the cosmos is a choice of the AI. If mass were the number of multiples accumulated since the beginning of the cosmos, it would grow by one part in 13.8 billion per year, $7.2\cdot10^{-11}$. With G, h and c fixed, the combination that lunar laser ranging measures ($G\,m^2/\hbar c$) would change by twice as much, $1.45\cdot10^{-10}$ per year, against $(-5.0 \pm 9.6)\cdot10^{-15}$: $1.5\cdot10^4$ standard deviations. **Outcome: excluded**, with the declared mapping (the full passage to the lunar-laser parameters is not derived here).

4.7 The readings that coincide with standard physics (M-a, M-c, M-e)

Mass as the Compton rhythm (M-a), as the slowing of rhythms around the body (M-c: the Komar mass of the metric of deposit 2.36.2 equals M at every radius) and as a value that, seen from far away, drops together with the rate of clocks (M-e: the Earth and the Moon fall in the same way, 0.88 standard deviations from the measurement) coincide with standard physics. They are identities or coincidences by construction: they are not independent checks of the thesis, but they show that the thesis is not contradicted at any of the points tested. In deposit 2.36.2, it should be recalled, the Schwarzschild metric comes from a declared postulate (the transfer postulate), not from the counts alone.

4.8 All outcomes

item	key number	outcome
M-0	—	reference
M-a	identity	compatible, equal to standard physics
M-b1	$\eta \approx -1.1\cdot10^{-42}$	not testable today; not conforming (the mass of a composite body is not shown)
M-b2	an angle that starts again after a full turn	not conforming

M-c	Komar mass = M at every radius	compatible, an identity of general relativity
M-d1	$2.7 \cdot 10^{11}$ σ; $1.3 \cdot 10^{10}$ times the limit; $1.7 \cdot 10^{4}$ σ	excluded
M-d2	—	compatible, equal to standard physics
M-e	0.88 σ	compatible, equal to standard physics
M-f1	no distinct state at rest	excluded by its own definition
M-f2	2 states per cycle (4 operations in Lloyd's count)	compatible
M-g	about 25 copies of difference	not concluded
M-h	$1.5 \cdot 10^{4}$ σ	excluded, with the declared mapping (an item of the AI)
N	Σ = 0.0591 eV	dated threshold registered

No item is distinctive (all read «contained» as a property of the body; neither is item M-i of §10, which uses the matrix of whoever observes).

4.9 Neutrinos: a threshold with a date

Deposit 2.38 read the lepton masses with the Koide relation and drew from it a prediction for the neutrinos, which is the same as Brannen's (2006). With the JUNO 2026 data the same relation gives masses of 0.00036, 0.00860 and 0.05009 eV and a sum of 0.0591 eV (between 0.0588 and 0.0594 eV when the JUNO data are varied within their errors), 0.37 meV above the minimum of the normal ordering. The rules, registered on 24 October 2026 with a deadline of 31 December 2035, are these:

- the reading falls if the inverted ordering is established at at least three standard deviations;
- the reading falls if a measurement of the sum (not a limit) places it above 0.0593 eV by more than three standard deviations;
- an upper limit at 99.7 % below 0.0587 eV would exclude the whole normal ordering with the lightest neutrino almost massless, not only this reading;
- a specific confirmation is not possible: the difference from the minimum, less than one thousandth of an electronvolt, is beyond the reach of foreseeable cosmology.

Today the DESI limit with the Feldman–Cousins method (0.053 eV) lies below both the prediction and the minimum of the normal ordering: the tension is between cosmology and neutrino oscillations, and does not concern this reading alone.

---

5. What is not claimed

- Mass is not derived, nor the value of any mass: the masses of the particles remain measured data.
- It is not claimed that the author's framework explains mass. What is claimed: the framework reads mass as the contained multiple; with the definition «the contained multiple is the value and remains a phase» the reading coincides with $E = mc^2$ and with relativity; the alternatives tested fall or remain open, as said in §4.
- The compatible items coincide with standard physics by identity or by construction: they are not independent proofs.
- It is not claimed that mass is a rhythm closed inside the object: this is the AI's reading of «contained». The author speaks of a multiple contained in the matrix of the point of observation; the translation using the matrix of whoever observes (§10) also coincides with standard physics.
- It is not claimed that the neutron fringes exclude copies of the Compton rhythm (§4.4).
- It is not claimed that the checkerboard predicts a violation of the equivalence principle: with the Planck step the effect is not testable, and the mass of a composite body on the checkerboard is not shown.
- The neutrino prediction is not new: it is Brannen's (2006), updated with the JUNO data.

- Not studied here: the fall on the checkerboard with copies arranged at random (as in causal sets: Johnston 2008; Dowker, Henson and Sorkin 2004); a rule that fixes the step of the checkerboard other than the Planck one; the mass threshold of the quantum world, treated in a separate work with a prediction dated to 2035.

6. The contact with observed objects

The twin deposit on observed objects (2.62) reads the object on which two observers agree as the common part of their records, made of copies taken at the same moment. In that static model the compatible translations of mass do not change the result: in them the contained multiple produces no distinct states, so it adds no copies to the observers' records; whether item M-g, not concluded, could change it remains open. If the open item (M-g) were true, the contained multiple would leave copies; in version 2.64 we had written that those copies would be an internal record of the particle and would not enter the observers' records without an interaction, but this is a hypothesis of the AI, not tested. The separation from the observed objects holds in the static model of deposit 2.62, with copies taken at the same moment; real copies accumulate in time, and whether the projection can be defined without the multiple remains open. In deposit 2.62 distance is read by triangulation; distance from luminosity needs the number of copies emitted by the source, which depends on the conditions of the source, among them the illumination and the temperature, and is not derived from the mass. For gravity the mass that counts is the value: the one that spreads according to Gauss's law is the total energy (M-d2).

7. How to reproduce

Three programs in Python 3 (with the sympy library), a few seconds in all on an ordinary computer: livello1_massa.py (the registered calculation), correzione_1.py (the numbers asked for by the first verification) and numeri_del_testo.py (the examples of §1 and §2). They need the AME2020 atomic-mass file (mass_1.mas20.txt) and the Particle Data Group constants. Every protocol and every program has a SHA-256 fingerprint and a UTC time registered before the calculation was started; the results are in JSON files. The protocol, the two corrections and the two independent verifications are in the package. Section 9 needs five more programs in Python 3 (with the numpy and scipy libraries), about three minutes in all: livello1_tempo.py (the registered calculation of the first test), diagnosi_V4_JR.py and controlli_giro1.py (the checks after the calculation and after the first verification), livello1b_registro.py (the test with the observer's record) and controlli_1b_giro1.py (its checks). They are in the folder 03_direzione_tempo, with the protocols, the outcome notes, the corrections and the four independent verifications. Section 10 needs two more programs in Python 3 (with the numpy library), a few seconds: mi_massa_osservatore.py and correzione_1_mi.py, in the folder 06_massa_matrice_osservatore with the protocol, the outcome notes, the corrections and the two independent verifications.

8. Provenance

The author is not a physicist. The translations, the programs and the calculations were made with an artificial intelligence (Claude, Anthropic) and verified by a second one (gpt-6-luna via Codex, OpenAI), in two rounds. In the first round ten remarks arrived: eight accepted, one accepted with a different argument, one in part. In the second round the remaining remarks were accepted, except one on the Brannen reference, which is not a journal article but a conference talk. Thanks to these verifications, two results were corrected: the copies of the Compton rhythm (from excluded to not concluded) and the whole body on the checkerboard (from excluded to not conforming). Three external reviews of the text then improved its clarity, and a verification of the package by the second AI corrected some sentences and the program that re-runs the calculations. The author's sentences from which the work arises are in the background file. In version 2.64.1 an independent check requested by the author (another instance of Claude, which did not know the conclusions of the first) found the undeclared translation choices corrected in this version; each point was rechecked against the original files. The corrections and section 9 were then verified by the second AI; the reports and the answers are in 04_controllo_262_264. An external review of the corrected version then added two clarifications, in §4.4 and §9.2. Item M-i of §10, suggested by the same review, was verified by the second AI in two rounds. The work on time of §9 was verified by the second AI in four rounds, two on the first test and two on the test with the observer's record.

---

9. Physical time and time with a direction

New section in version 2.64.1. It can be read on its own.

The previous sections treat time as the read multiple: a row of distinct states, like the ticks of a clock. This is **physical time**, the time that clocks measure. A row of ticks, however, does not say by itself which way it goes: the laws of microscopic physics work almost in the same way forwards and backwards, yet we remember the past and not the future,

and heat passes from hot to cold. This way is **time with a direction**.

The author's framework gives a reading of it: the direction of time is the increase in the number of relations, linked to one another in a network, inside the matrix of whoever observes (the exact words are in the background file). This section says what physics knows about the direction of time, how we translated the author's sentence and what comes out of the comparison.

9.1 What is already known

- **Almost symmetric laws, a world with a direction.** Microscopic laws do not distinguish past from future, except for a small violation in weak interactions, measured directly with B mesons (BaBar 2012). The direction we see comes, in the most widespread explanation, from an initial condition: the universe started in a very ordered state, and since then entropy (a measure of disorder) has grown (Lebowitz 1996).
- **Heat can flow backwards.** Two bodies that start each at its own temperature and *without correlations* between them and exchange energy only with each other in a process that conserves total energy and entropy, exchange heat only from hot to cold. If instead they start correlated (the properties of one are not independent of those of the other), heat can pass from cold to hot, but only by spending correlation: the sum of the heats received, each divided by the temperature of its own body, cannot fall below the change of their mutual information, so a backward flow requires the mutual information to decrease (Partovi 2008; Jennings and Rudolph 2010). **Mutual information** says, in bits, how much knowing the state of one body tells about the state of the other; one bit is the information of a yes-or-no answer.
- **The measurements.** Micadei and others (2019) observed it with two nuclear spins of a molecule: heat passed from the colder to the hotter for about two thousandths of a second, until the initial correlation was spent. Naghdi and others (2022) obtained it in a chain of three qubits on a quantum computer; Huang and others (2026) in a system of four spins, where, for the same correlation, the direction of heat depends on the phase of the coherence.
- **The direction of heat is not the whole direction of time.** Latune and others (2019) show cases in which heat flows backwards while entropy production stays positive: for them the arrow of time is not reversed. Jennings and Rudolph (2010, in their comment on Maccone) give an example in which local entropy decreases while memory grows. Hartle (2005): in the ideal model a record forms without a necessary increase of total entropy; it is erasure that requires it. Wolpert and Kipper (2023): the memories they call Type 3, which start from a «ready» state, need an initialisation that increases global entropy.
- **Complexity, growth, point of view.** The complexity of a quantum state (how many elementary operations are needed to prepare it) grows for a long time like an entropy (Brown and Susskind 2018), at a maximum rate fixed by the energy (Brown and others 2016, starting from the Margolus–Levitin bound of §1). In causal sets the passage of time is the birth of new elements (Rideout and Sorkin 2000; Dowker 2014). Rovelli (2015) proposes that the arrow depends on how the observer groups states; Caticha (2011) builds an «entropic» time, made of inference steps, that has a built-in direction.

In short: known physics already distinguishes the direction of heat from other arrows (that of memory, of complexity, of growth), and ties the direction of heat to the correlations between the observed parts.

9.2 How we tested the author's sentence

We translated «increase in the number of relations linked in a network» in twelve ways, plus a reference (known physics), and registered them before the calculations: for example the number of pairs of parts correlated above a threshold, the total correlation of the network counted in bits, the sum of the mutual informations of the pairs, the relations along direct links only, the redundancy of copies, complexity, the growth of the row of events, the ratio between two rhythms of time, the loop of deposit 2.48, the number of readings compared with the information carried by each. Then we compared them with three numerical benches reconstructed from published experiments and papers, with an exact calculation and no fitted parameters: the two spins of Micadei (the program recovers the backward heat up to 2.199 ms, against about 2.1 ms in the paper), the three-qubit chain of Naghdi (recovered in its qualitative features) and the three qubits of Jennings and Rudolph (here the program does not recover the point of the paper's example in which the coldest part gives up heat; it recovers it by inverting the relative phase between the initial correlation and the interactions, which, however, is a physically different model. The reason for the difference with the paper was not found. The counts of this bench hold for the registered reconstruction; the main result of §9.3 does not depend on them: the first kind of opposite moments occurs in the benches of Micadei and Naghdi, the second in that of Naghdi).

A correction, stated at once. In the first test we had added a criterion of our own: a quantity reads the direction of time only if it decreases every time heat flows backwards. This link between direction and heat is not in the author's sentence and nobody has tested it; the author observed that the direction of heat could in some cases be contrary to the complication of the matrix. The criterion was removed: two readings that were «excluded» by it (the number of pairs above a threshold and the relations along direct links only) no longer are. Moreover, the first test counted the relations

between the observed parts, not those inside the matrix of whoever observes. For this reason we registered a second test, with an **observer's record**: a few memory cells that start empty, with no energy of their own, and copy the energy level of an observed part. The **complication** is the number of relations, in bits, that the record carries with the observed parts (the mutual information between the record and the parts); with several cells the correlation among the memories is also measured, separately, and it does not enter the comparisons with the direction of heat. In two benches we took one of the parts itself as the observer's memory.

9.3 What comes out

The relations between the observed parts follow the heat. Counted in bits over the whole network (the **total correlation**), the correlations between the observed parts decrease every time heat flows backwards: this is an exact identity, which the program checks to within $5.2 \cdot 10^{-14}$, and it is the Jennings–Rudolph bound extended to several parts. The converse does not hold: correlation can also decrease when heat flows in the normal direction. This is known physics, and it concerns the observed parts, not the matrix of whoever observes. The sum of the mutual informations of the pairs also decreases at every reversal, in the seven cases tested. The number of correlated pairs above a threshold and the relations along direct links only, instead, do not follow the heat: for every threshold tested there is a stretch in which heat flows backwards and the number of pairs does not decrease, and in the Jennings–Rudolph bench the relations along direct links stay at zero while heat flows backwards. These two readings would need a threshold fixed by an independent criterion, which today does not exist.

The relations in the observer's record do not always go with the heat. In every bench tested there are moments with opposite signs with respect to the initial values: backward heat with the complication above its initial value, or heat in the normal direction with the complication below it.

- **Two spins of Micadei, one reading.** In all the 2198 instants in which heat flows backwards, the record, which started empty, carries between 0.365 and 1.166 bits of relations, and the reading does not change the heat. It is almost obvious by construction: it says that nothing prevents the record from growing while heat flows backwards.
- **Four readings during the process.** The record rises at each reading: 0.73, 1.04, 1.24, 1.38 bits. The readings reduce the amount of backward heat (at 1.0 ms −0.41 peV against −0.71 without readings; at 1.5 ms −0.30 against −0.61) but make it last until the end of the computed time. We had registered the expectation that after the first reading the backward heat would stop: it did not.
- **Three qubits of Jennings and Rudolph, with one of the parts as memory.** Here the observer's memory is itself the correlation that heat spends: at all the 1366 points where heat flows backwards, the complication is below its initial value of 0.520 bits. The registered expectation (at least one point with backward heat and complication above the initial value) did not come true. There are, however, 12 754 points of the other kind: heat in the normal direction and complication below the initial value (125 points are not classified, because the changes are within the tolerance).
- **Chain of Naghdi, with one of the parts as memory.** In one case, 60 instants with backward heat and complication above the initial value; in another case none, but 931 instants with normal heat and complication below (in each case one instant is not classified, because it is within the tolerance).

What seems to decide: when the observer writes new records, its complication rises whatever the direction of heat; when its memory is the very correlation that heat spends, it is below the initial value every time heat flows backwards. The author's sentence, that the direction of heat could in some cases be contrary to the complication of the matrix, therefore has examples in standard quantum mechanics. It holds for these models and this way of counting, not as a general law.

The number gives a bound, the phase chooses the direction. Two initial states of the two spins of Micadei with the same temperatures, the same mutual information and the same counts give heat in opposite directions in the first 0.2 ms: no quantity of number or amount fixes by itself the local direction of heat; it is chosen by the phase of the correlation, as in the experiment of Huang and others.

The other readings. Complexity is not measured today, so it is not testable. The growth of the row of events is compatible as a reading of the passage of time. The number of readings grows by one at each reading, while the information carried by one reading falls as 1/T: the program recovers the known asymptotic value to within 0.01 %, and the «information time» grows as the logarithm of the number of readings; this is a known result of Bayesian statistics, with the precedent of Caticha.

reading	outcome
known physics (reference); coarse-grained entropy	compatible

correlated pairs above a threshold	does not follow the heat; not conforming until the threshold is fixed by an independent criterion
total correlation between the observed parts, in bits	follows the heat (exact identity); compatible, not distinctive
sum of the mutual informations of the pairs	follows the heat in the cases tested; compatible, not distinctive
relations along direct links only	does not follow the heat; not conforming until the threshold is fixed by an independent criterion
redundancy of copies	compatible, not distinctive
complexity	not testable today
growth of the row of events	compatible as the passage of time
two rhythms of time at the same point	compatible if the ratio is the same for every process; not verified directly
the loop (deposit 2.48)	compatible, not distinctive
the observer's network (the part of the network that contains the record)	compatible, not distinctive
number of readings and information per reading	compatible, not distinctive (precedent: Caticha 2011)
complication of the observer's record	**compatible, not distinctive; it does not always go with the heat: in all the benches tested there are moments with opposite signs with respect to the initial values**

No reading gives today a prediction different from known physics. Read as the complication of the observer's record, the direction of time is a quantity different from the direction of heat: in agreement at some moments, opposite at others.

9.4 The contact with mass

In §4.3 a mass at rest, in a state of definite energy, produces no distinct states: its state changes only by a phase. As long as it stays isolated, therefore, its complexity does not grow and it brings no new relations with a record; when it interacts with a record it can become correlated with it, like any system. Reading this as «the contained multiple does not contribute to the direction of time, which arises when something is read» is an interpretive hypothesis of the AI, which the benches of this section do not test. The maximum rate at which complexity can grow is fixed by the energy, through the same Margolus–Levitin bound that in §4.3 is the bridge between mass and time: this is a consistency between the two parts of the work, not a proof. Unlike the eleven translations of mass of §3, the direction of time was translated using the observer's record, as item M-i of §10 is.

9.5 What is not claimed

- The direction of time is not derived: it is read, and the reading is compatible with known physics.
- It is not claimed that the complication of the record is always contrary to the heat, nor that it usually is: at some moments the signs agree, at others they are opposite.
- The observer is a minimal model and the benches are small; the complication is counted in bits, and other ways of counting (for example complexity) were not tested.
- Two registered expectations did not come true, one validation was not reproduced and one registered control formula was written wrongly (the right one holds exactly): all of this is reported in the outcome notes of the folder 03_direzione_tempo.

---

10. Mass in the matrix of whoever observes (item M-i)

New section in version 2.64.1. The eleven translations of §3 read «contained» as a rhythm closed inside the observed body; the author speaks of a multiple «contained directly in the matrix of the point of observation». This section writes and tests the missing translation, pointed out by an external review.

The translation. The observer copies the same body at each tick of its own clock. The row of copies divides into two parts: what changes from one copy to the next, which can be seen only by comparing two copies (the read multiple, time), and what stays the same in all the copies, which can be read from a single copy (the contained multiple). Mass is the size of this second part. Its value is the rhythm that the observer's time gives to a contained block, and it exists only if the block has a single rhythm (technically, if it is an eigenspace of the generator of time: a clarification written after the first calculations). What observers in different motions have in common, in the sense of the common part of deposit 2.62 (Gács and Körner 1973), is mass as we usually understand it. In the model the ticks follow the ordinary time of quantum mechanics: no internal clock is built. Protocol, programs and notes were registered before the calculations and verified by the second AI in two rounds (folder 06_massa_matrice_osservatore). The choices that tie the author's words to the calculated quantities are the AI's and are listed in the outcome note.

1. What is conserved stays contained. In a test body made of two bound pieces, the copies of the total energy stay identical at every distance between ticks: everything is contained (1.747 bits). The copies of the levels of the single pieces keep only the number of excitations (1.271 bits), a count, which is contained too; the copies of one piece alone keep nothing. This is what is known from non-demolition measurements (Braginsky, Vorontsov and Thorne 1980): conserved quantities give the same outcome again.

2. The value weighs, binding included. The bound block has value 0.7197 (in model units) against a mean count of the pieces of 0.8445: the value is lower by 0.1248, that is, by the binding energy g^2/Ω, as a nucleus weighs less than its pieces. The count is as contained as the value, but it does not have a single rhythm: what selects the value is the rule that assigns a value only to blocks with a single rhythm, a choice of the AI. With the opposite choice, which counts the copies of the pieces, the item does not follow from the data: if it were adopted, it would give the numbers that §4.2 excludes (MICROSCOPE: $2.7 \cdot 10^{11}$ standard deviations). What stays contained, and which blocks receive a value, depends on what the observer copies and on how it ticks: copying one piece alone, no block receives a value; with ticks at a fixed rate (stroboscopic), more seems contained, but the additional blocks receive no value.

3. What observers in motion have in common distinguishes the masses. Two observers with different velocities record the energy and momentum of five particles (electron, muon, proton, neutron, deuteron). With recording cells of 0.01 and 0.1 MeV the common part of their records separates the five particles, and each common component contains the invariant mass: for the proton, with 0.01 MeV cells, the interval runs from 938.245 to 938.299 MeV. With 1 and 10 MeV cells proton and neutron merge; with energy alone they merge at every resolution. The energy of a photon is not common to the observers and its invariant is zero; two opposite photons closed in a box give an invariant of 2 MeV. This is mass as the invariant of relativity (Wigner 1939), read as the common part of the records: this reading is the AI's, not Wigner's.

4. Complication and mass of the record are different quantities. In the record of section 9, with cells that have no energy of their own, the four readings carry 0.731, 1.044, 1.237 and 1.376 bits of relations and no energy. If the cells have an energy gap ($h \cdot 1$ kHz), the energy of the record rises to 2.75 peV with the same relations; four excited cells that read nothing carry 16.5 peV and zero bits. In the model the mass of the record follows its energy, not its complication.

5. In no case examined does the item give a number different from standard physics. For clocks in gravity it coincides with M-e (general relativity), for observers in motion with special relativity; for atoms in a superposition of two internal states the result of Rosi and others (2017) is compatible with zero at 1.1 standard deviations, and the item gives no prediction different from the standard one for this comparison. Where the observer copies no conserved quantity with a single rhythm, the item gives no value: it is not conforming, not distinctive. A nearby reading, mass as information multiplied by $k_B T \ln 2 / c^2$ (Vopson's proposal, 2019), introduces a constant and a temperature (it does not respect the criterion on new constants); in the record without an energy gap it would give mass without energy, so it does not follow the criterion $E = mc^2$ (this depends on the choice that ties bits to mass); and it is not testable today in the cases of the protocol. The quantum phase of free fall in gravity was not compared.

item	what it says	outcome
M-i1	mass = the part of the copies that stays the same at every tick; value = rhythm, only for blocks with a single rhythm; common to observers in motion	compatible, equal to standard physics by construction; not conforming where no copied block has a single rhythm (for example copying one piece alone)
M-i2	as M-i1, with value E/c^2 of the single observer	does not follow the limiting case of light (depends on a choice of the AI)
M-i3	as M-i1, with value = contained copies of the pieces	does not follow from the data (MICROSCOPE, as M-d1; depends on a choice of the AI)
M-i4	mass = energy of the record	compatible as an identity of definition in the model

M-i5	(an item of the AI) mass = complication in bits × k_B T ln 2 / c^2 (Vopson 2019)	does not respect the criterion on new constants; $E = mc^2$ does not follow in the record without an energy gap (depends on the choice that ties bits to mass); not testable today in the cases tested

What it adds. It translates the author's words, the matrix of whoever observes, and not a property of the body, and it coincides with $E = mc^2$ and with relativity within the measurements used in §4. The author's distinction between read multiple and contained multiple corresponds, in the record, to the division of the copies into what changes and what stays: a consistency with non-demolition measurements and with the Page–Wootters framework (1983), in which time arises from correlations with a clock (illustrated in the laboratory by Moreva and others 2014; developed in theory by Giovannetti, Lloyd and Maccone 2015). The model does not test that mechanism. The common part of deposit 2.62, applied to the copies of a single observer in time, gives what is conserved; applied to observers in motion, it distinguishes the masses. Prior work to be cited first: de Broglie (1924), who associated with mass a periodic phenomenon with $h\nu = mc^2$; Noether (1918), for the link between energy and symmetry in time; Wigner (1939), for mass as an invariant.

What is not claimed. That MM derives mass. That mass depends on the observer: what depends on the observer is what stays contained in its copies, and the energy in its frame, as in relativity. That the value does not depend on how one copies: where it is assigned it is, by construction of the rule, the rhythm of the block. A new prediction.

Glossary

- **Backward heat:** heat passing from the colder body to the hotter one; it is possible only if the bodies start correlated, and it is limited by the correlation that is spent.
- **Binding energy:** the energy that must be given to a body to separate its pieces.
- **Bit:** the unit of information: that of a yes-or-no answer.
- **Causal set:** a model in which space-time is made of discrete elements linked by relations of before and after; the passage of time is the birth of new elements.
- **Complexity (of a quantum state):** the minimum number of elementary operations needed to prepare it.
- **Complication (of the observer's record):** in §9, the number of relations, counted in bits, that the record of whoever observes carries with the observed parts; with several cells the correlation among the memories is also measured, separately.
- **Compton frequency:** the rhythm of a mass, $\nu = mc^2/h$.
- **Contained (contained multiple):** for the author, the multiple contained directly in the matrix of the point of observation; in the AI's reading used in this work, the counting that stays inside the thing as a rhythm, without producing distinct states; it is mass. In §10 (item M-i), the part of the observer's copies that stays the same at every tick.
- **Copy:** in the author's framework, a repetition of the common unit; in experiments, a recorded mark, in an internal degree of freedom of the particle or in the environment (for example a scattered photon), that carries the imprint of something: the value of an object (as in deposit 2.62) or the path taken (as in §4.4).
- **Count of the pieces:** the sum of the masses of the pieces of a body taken one by one, without the binding.
- **Distinct (orthogonal) states:** states that a measurement can distinguish without error.
- **Eigenstate (of energy):** a state with definite energy; in time it changes only by a phase, while nothing observable changes.
- **Entropy:** a measure of disorder, or of the information missing to know the state; in §9 it is counted in bits.
- **Equivalence principle:** all bodies fall in the same way, whatever their composition.
- **Feldman–Cousins (method):** a way of computing a limit that takes into account that a quantity cannot be negative.
- **Fringes (contrast of):** the light and dark stripes of interference; the contrast says how sharp they are.
- **GCD (greatest common divisor):** in the author's framework, the unit that two entities have in common and that is counted in each.
- **Gravitational redshift:** the slowing of rhythms near a mass, seen from far away.
- **Interferometer:** an instrument that splits a particle (or a wave) into two paths and recombines them, to see the fringes.
- **Invariant mass:** the mass that observers in any motion find equal, obtained from energy and momentum; in §10, what their records have in common.
- **Komar (mass):** the mass of a body read from the way it slows the clocks around it, summed over a surface that encloses it.
- **Lunar laser ranging:** the measurement of the distance of the Moon with laser pulses reflected by mirrors left on the Moon.
- **Margolus–Levitin (bound):** the maximum rate at which a system of given energy can pass through distinct states.
- **Mass at rest, inertial, gravitational:** mass as the energy of the body at rest; as resistance to a push; as what is

attracted (passive) or what attracts (active). The equivalence principle says that inertial and passive gravitational mass are the same, so that all bodies fall in the same way; the equality with active mass is checked separately (for example by lunar laser ranging).
- **Mass defect:** the fact that a bound body weighs less than its separate pieces.
- **Matrix:** in the author's framework, the set of copies of the common unit, counted from a point of view.
- **MICROSCOPE:** a satellite that compared the fall of two cylinders of titanium and platinum in orbit.
- **Multiple:** in the author's framework, counting the common unit forward, one copy after another.
- **Mutual information:** how much knowing the state of one part tells about the state of another, counted in bits.
- **Neutrino ordering:** which of the three neutrinos is the lightest; «normal» if the lightest is the first.
- **Non-demolition measurement:** a measurement that, when repeated, gives the same outcome again, because the measured quantity is conserved.
- **Nordtvedt (effect):** the hypothesis that bodies with much gravitational binding energy fall differently; lunar laser ranging excludes it.
- **Observer's record:** in §9, a few memory cells that start empty and copy the state of an observed part; it is the AI's translation of the matrix of whoever observes.
- **Phase:** the angle of a rhythm; a phase that changes in the same way across the whole state is not observable on its own.
- **Physical time:** the time measured by clocks; in the thesis, the read multiple.
- **Planck (mass, time):** the scales built from h, c and G: about $2.2 \cdot 10^{-8}$ kg and $5.4 \cdot 10^{-44}$ s.
- **Proper time:** the time marked by a clock travelling with the body.
- **Read (read multiple):** in the thesis, the counting that becomes a row of distinct states; it is time.
- **Standard deviation (σ):** the measure of the uncertainty of a datum; a result three standard deviations away from a prediction casts strong doubt on it, 10^{11} away excludes it.
- **Step (of the checkerboard):** the interval of time and space between two squares.
- **Time with a direction:** the direction of time: we remember the past and not the future, and heat passes from hot to cold.
- **Total correlation:** the sum of the entropies of the parts minus the entropy of the whole: how many relations, in bits, link the parts of a network.
- **Translation:** in this work, a mathematical definition of the thesis that allows a calculation comparable with the data.
- **Value:** in the thesis, the total energy contained in a thing, binding included; divided by c^2 it is its mass.
- **Walker:** on the checkerboard, the particle that zigzags along the directions of light rays.

Package contents

- README.md (this text); LEGGIMI_IT.md (the same text in Italian); AUTHOR_BACKGROUND.md (the author's sentences, kept apart from the main text).
- 02_massa/LIVELLO1_PROTOCOLLO_v1.md: the thirteen items (eleven translations, the reference M-0 and the neutrino threshold N), the data, the seven criteria and the outcome rules, registered before any calculation (in Italian, as registered).
- 02_massa/CORREZIONE_1_LIVELLO1.md and CORREZIONE_2_LIVELLO1.md: the answers to the two verifications, each registered before its program; NOTA_ESITO_LIVELLO1_v1.md, v2, v3: the outcome notes (v3 is the final one).
- 02_massa/programmi: the three programs; 02_massa/risultati: their JSON results and logs.
- 02_massa/VERIFICA_ASTRA_G1 and VERIFICA_ASTRA_G2: the two verification reports on the calculations (gpt-6-luna via Codex); VERIFICA_PACCHETTO: the report of the same verifier on the package and the answers to its remarks.
- 03_direzione_tempo (section 9): LIVELLO1_PROTOCOLLO_v1.md and LIVELLO1B_PROTOCOLLO_REGISTRO_v1.md, the protocols of the two tests, each registered before its calculations; NOTA_ESITO_LIVELLO1_v1.md to v6 and NOTA_ESITO_LIVELLO1B_v1.md to v3, the outcome notes (v6 and v3 are the final ones); CORREZIONI_GIRO1, CORREZIONI_GIRO2, CORREZIONI_1B_GIRO1 and CORREZIONI_1B_GIRO2, the answers to the verifications; VERIFICA_ASTRA_G1, VERIFICA_ASTRA_G2, VERIFICA_ASTRA_1B_G1 and VERIFICA_ASTRA_1B_G2, the four verification reports; programmi, the five programs; risultati, their JSON results and logs (all in Italian, as registered). The Level-0 inventory of the 52 sources, the source cards, the plan and the verification prompts stay outside the archive (fingerprints in the REGISTRO.txt files).
- 06_massa_matrice_osservatore (section 10): PROTOCOLLO_M_i_v1.md, the protocol registered before the calculations; NOTA_ESITO_M_i_v1.md to v3, the outcome notes (v3 is the final one); CORREZIONE_1_M_i.md and CORREZIONE_2_M_i.md, the answers to the two verifications; VERIFICA_ASTRA_G1 and VERIFICA_ASTRA_G2, the two verification reports; programmi, the two programs; risultati, their JSON results and logs (in Italian, as registered).
- 04_controllo_262_264/PIANO_CORREZIONI_v1.md: the plan of the corrections of version 2.64.1, registered before the texts were changed (in Italian); 04_controllo_262_264/REGISTRO.txt: fingerprints of the independent check and of its verification, which stay outside the archive because they quote the author's working

sentences.
- 04_controllo_262_264/VERIFICA_ASTRA_264_1_G1 and following rounds: the reports of the second AI on the corrections and on section 9 of version 2.64.1, and the answers (in Italian).
- REGISTRO.txt in the main folder and in each subfolder: SHA-256 fingerprint and UTC time of every registration. A few registered working files are not in the archive because they quote the author's working sentences (the premises, the plan, the Level-0 inventory, the list of sources, the verification prompts, which contain the whole material); their fingerprints are in the REGISTRO.txt files, and they can be shown on request.
- Data: the AME2020 atomic-mass table (mass_1.mas20.txt, IAEA Atomic Mass Data Center) is not redistributed; run_all.py checks its fingerprint. The constants of the Particle Data Group (2025) are written in the programs. Third-party papers are not redistributed; they are listed in the References.
- run_all.py: re-runs every calculation and compares the results with the registered ones; IMPRONTE.sha256: fingerprints of all the other files in the archive.
- LICENSE.md: texts under CC BY 4.0, code under the MIT licence.
- In the working records this deposit is called «massa come multiplo, Livello 1» (folder MM_sviluppo_INDIGERIBILI/02_massa); file names and registered documents keep their Italian names and language.

How to run

python run_all.py mass_1.mas20.txt

The script checks the fingerprint of the AME2020 file, runs the ten programs (three on mass, five on the direction of time, two on item M-i) into a temporary folder and compares every number with the registered JSON files (relative tolerance 10^{-9} of the larger value, with no absolute floor). A few registered numbers are rounding residuals of a quantity that is zero by construction (the residual of an exact identity, the initial correlation of parts prepared uncorrelated, a zero of a series): their last digits change from one run to another, even on the same machine. For these paths only, listed in the script, two values that fail the relative test are accepted if both are below a stated bound (the registered tolerance, 10^{-9} or 10^{-12}, for the two items that have one; 10^{-13}, a hundred times the rounding error of double precision on these quantities, for numbers that are zero by construction), and they are all printed as residuals; the position where a numerical optimiser stops on a function that is zero on a whole region is not compared. It ends with «ALL MATCH». Tested with Python 3.13.13, sympy 1.14.0, numpy 2.4.6 and scipy 1.18.0; about three minutes.

Related records

- 2.62, the twin deposit on observed objects (DOI 10.5281/zenodo.23609967): the other half of the same step of the framework; corrected in version 2.62.1.
- 2.25, in record 2.27 (DOI 10.5281/zenodo.22667516): the Compton frequency under the postulate $\varphi \equiv S/\hbar$ («frozen time»).
- 2.36.2 (DOI 10.5281/zenodo.23287965): premise P4 (the number of copies is mass) and the metric used for the Komar mass.
- 2.38 (DOI 10.5281/zenodo.23055978): the Koide relation and the neutrino prediction updated here.
- 2.40 (DOI 10.5281/zenodo.23280168): the checkerboard, mass as turning angle and the mass-dependent fall.
- 2.48 (DOI 10.5281/zenodo.23352814): the loop, one of the readings of the direction of time tested in §9.

References

- P. Arrighi, V. Nesme, M. Forets, The Dirac equation as a quantum walk: higher dimensions, observational convergence, J. Phys. A 47, 465302 (2014), arXiv:1307.3524.
- BaBar Collaboration (J. P. Lees and others), Observation of time-reversal violation in the B0 meson system, Phys. Rev. Lett. 109, 211801 (2012), arXiv:1207.5832.
- P. L. Bartlett, D. Van Buren, Equivalence of active and passive gravitational mass using the Moon, Phys. Rev. Lett. 57, 21 (1986).
- I. Bialynicki-Birula, Weyl, Dirac, and Maxwell equations on a lattice as unitary cellular automata, Phys. Rev. D 49, 6920 (1994).
- V. B. Braginsky, Y. I. Vorontsov, K. S. Thorne, Quantum nondemolition measurements, Science 209, 547-557 (1980).
- C. Brannen, Koide's mass formula for neutrinos, APS Northwest Section Meeting (2006); talk «Koide formula for neutrino masses», Joint Meeting of Pacific Region Particle Physics Communities, Waikiki, 1 November 2006.
- L. de Broglie, Recherches sur la théorie des quanta, tesi, Parigi (1924); Annales de Physique 3, 22 (1925).
- A. R. Brown, D. A. Roberts, L. Susskind, B. Swingle, Y. Zhao, Complexity, action, and black holes, Phys. Rev. D 93, 086006 (2016), arXiv:1512.04993.

- A. R. Brown, L. Susskind, Second law of quantum complexity, Phys. Rev. D 97, 086015 (2018), arXiv:1701.01107.
- A. Caticha, Entropic dynamics, time and quantum theory, J. Phys. A 44, 225303 (2011), arXiv:1005.2357.
- R. Colella, A. W. Overhauser, S. A. Werner, Observation of gravitationally induced quantum interference, Phys. Rev. Lett. 34, 1472-1474 (1975).
- T. Damour, Experimental tests of gravitational theory, in Particle Data Group, Review of Particle Physics, revision of August 2025.
- G. M. D'Ariano, P. Perinotti, Derivation of the Dirac equation from principles of information processing, Phys. Rev. A 90, 062106 (2014), arXiv:1306.1934.
- F. Dowker, The birth of spacetime atoms as the passage of time, Annals of the New York Academy of Sciences (2014), DOI 10.1111/nyas.12542, arXiv:1405.3492.
- F. Dowker, J. Henson, R. D. Sorkin, Quantum gravity phenomenology, Lorentz invariance and discreteness, Mod. Phys. Lett. A 19, 1829-1840 (2004), arXiv:gr-qc/0311055.
- S. Dürr and others, Ab initio determination of light hadron masses, Science 322, 1224 (2008), arXiv:0906.3599.
- W. Elbers and others (DESI), Constraints on neutrino physics from DESI DR2 BAO and DR1 full shape, arXiv:2503.14744 (2025).
- N. Gaaloul and others, STE-QUEST: Space Time Explorer and QUantum Equivalence principle Space Test, arXiv:2211.15412.
- P. Gács, J. Körner, Common information is far less than mutual information, Problems of Control and Information Theory 2, 149-162 (1973).
- V. Giovannetti, S. Lloyd, L. Maccone, Quantum time, Phys. Rev. D 92, 045033 (2015).
- D. M. Greenberger, Conceptual problems related to time and mass in quantum theory, arXiv:1011.3709 (2010).
- J. B. Hartle, The physics of «now», Am. J. Phys. 73, 101-109 (2005), arXiv:gr-qc/0403001.
- K. Huang and others, Reversing heat flow by coherence in a multipartite quantum system, arXiv:2601.00198 (2026).
- D. Jennings, T. Rudolph, Entanglement and the thermodynamic arrow of time, Phys. Rev. E 81, 061130 (2010), arXiv:1002.0314.
- D. Jennings, T. Rudolph, Comment on «Quantum resolution to the arrow of time dilemma», Phys. Rev. Lett. 104, 148901 (2010), arXiv:0909.1726.
- S. Johnston, Particle propagators on discrete spacetime, Class. Quantum Grav. 25, 202001 (2008), arXiv:0806.3083.
- JUNO Collaboration, Measurement of neutrino oscillation parameters at JUNO and indication of the neutrino mass ordering, arXiv:2609.37895 (2026).
- Y. Koide, Lett. Nuovo Cim. 34, 201 (1982); Phys. Lett. B 120, 161 (1983).
- A. Komar, Covariant conservation laws in general relativity, Phys. Rev. 113, 934 (1959).
- S.-Y. Lan and others, A clock directly linking time to a particle's mass, Science 339, 554 (2013).
- C. L. Latune and others, Heat flow reversals without reversing the arrow of time: the role of internal quantum coherences and correlations, Phys. Rev. Research 1, 033097 (2019), arXiv:1907.10615.
- J. L. Lebowitz, Microscopic reversibility and macroscopic behavior: physical explanations and mathematical derivations, arXiv:cond-mat/9605183 (1996).
- S. Lloyd, Ultimate physical limits to computation, Nature 406, 1047 (2000), arXiv:quant-ph/9908043.
- Y. Margalit and others, A self-interfering clock as a «which path» witness, Science 349, 1205 (2015), arXiv:1505.05765.
- N. Margolus, L. B. Levitin, The maximum speed of dynamical evolution, Physica D 120, 188-195 (1998), arXiv:quant-ph/9710043.
- D. A. Meyer, From quantum cellular automata to quantum lattice gases, J. Stat. Phys. 85, 551 (1996), arXiv:quant-ph/9604003.
- K. Micadei and others, Reversing the direction of heat flow using quantum correlations, Nature Communications 10, 2456 (2019), arXiv:1711.03323.
- E. Moreva and others, Time from quantum entanglement: an experimental illustration, Phys. Rev. A 89, 052122 (2014).
- H. Müller, A. Peters, S. Chu, A precision measurement of the gravitational redshift by the interference of matter waves, Nature 463, 926 (2010).
- S. Naghdi and others, Inducing heat reversal in a three-qubit spin chain, arXiv:2205.09300 (2022).
- E. Noether, Invariante Variationsprobleme, Nachrichten der Gesellschaft der Wissenschaften zu Göttingen, Math.-Phys. Klasse, 235-257 (1918).
- K. Nordtvedt, Phys. Rev. 170, 1186 (1968).
- D. N. Page, W. K. Wootters, Evolution without evolution: dynamics described by stationary observables, Phys. Rev. D 27, 2885-2892 (1983).
- Particle Data Group, Review of Particle Physics, 2025 update: physical constants.
- M. H. Partovi, Verschraenkung versus Stosszahlansatz: disappearance of the thermodynamic arrow in a high-correlation environment, Phys. Rev. E 77, 021110 (2008), arXiv:0708.2515.
- I. Pikovski, M. Zych, F. Costa, Č. Brukner, Universal decoherence due to gravitational time dilation, Nature Physics 11, 668-672 (2015), arXiv:1311.1095.

- D. P. Rideout, R. D. Sorkin, A classical sequential growth dynamics for causal sets, Phys. Rev. D 61, 024002 (2000), arXiv:gr-qc/9904062.
- G. Rosi and others, Quantum test of the equivalence principle for atoms in superpositions of internal energy eigenstates, Nature Communications 8, 15529 (2017), arXiv:1704.02296.
- A. Roura, Gravitational redshift in quantum-clock interferometry, Phys. Rev. X 10, 021014 (2020), arXiv:1810.06744.
- C. Rovelli, Is time's arrow perspectival?, arXiv:1505.01125 (2015).
- V. V. Singh, J. Müller, L. Biskupek, E. Hackmann, C. Lämmerzahl, Equivalence of active and passive gravitational mass tested with lunar laser ranging, Phys. Rev. Lett. 131, 021401 (2023), arXiv:2212.09407.
- C. Struckmann and others, Platform and environment requirements of a satellite quantum test of the weak equivalence principle at the 10^{-17} level, Phys. Rev. D 109, 064010 (2024), arXiv:2310.04212.
- P. Touboul and others, MICROSCOPE mission: final results of the test of the equivalence principle, Phys. Rev. Lett. 129, 121102 (2022), arXiv:2209.15487.
- M. M. Vopson, The mass-energy-information equivalence principle, AIP Advances 9 (2019), DOI 10.1063/1.5123794.
- M. Wang, W. J. Huang, F. G. Kondev, G. Audi, S. Naimi, The AME2020 atomic mass evaluation (II), Chinese Phys. C 45, 030003 (2021).
- E. Wigner, On unitary representations of the inhomogeneous Lorentz group, Annals of Mathematics 40, 149-204 (1939).
- P. Wolf, L. Blanchet, C. J. Bordé, S. Reynaud, C. Salomon, C. Cohen-Tannoudji, Does an atom interferometer test the gravitational redshift at the Compton frequency?, Class. Quantum Grav. 28, 145017 (2011), arXiv:1012.1194.
- D. H. Wolpert, J. Kipper, Memory systems, the epistemic arrow of time, and the second law, arXiv:2309.10927 (2023).
- M. Zawisky, M. Baron, R. Loidl, H. Rauch, Testing the world's largest monolithic perfect crystal neutron interferometer, Nucl. Instrum. Methods Phys. Res. A 481, 406-413 (2002).
- M. Zych, F. Costa, I. Pikovski, Č. Brukner, Quantum interferometric visibility as a witness of general relativistic proper time, Nature Communications 2, 505 (2011), arXiv:1105.4531.
- M. Zych, Č. Brukner, Quantum formulation of the Einstein equivalence principle, Nature Physics 14, 1027 (2018), arXiv:1502.00971.
- Earlier deposits of the author cited: 2.25, in record 2.27 (DOI 10.5281/zenodo.22667516); 2.36.2 (DOI 10.5281/zenodo.23287965); 2.38 (DOI 10.5281/zenodo.23055978); 2.40 (DOI 10.5281/zenodo.23280168); 2.48 (DOI 10.5281/zenodo.23352814); 2.62, on observed objects (DOI 10.5281/zenodo.23609967), corrected in version 2.62.1.